Provider Manager Application Backend Guide

This document describes how the backend of the Provider Manager application works. It also explains how to interact with the system.

1. Overview

- The backend is a C# ASP.NET Core 8 API.
- The system uses Entity Framework Core and an SQLite database to store and manage data.
- The backend connects to the React frontend.
- The API provides RESTful endpoints to manage providers and licenses.

2. Database Schema

The database contains two tables: the Providers table and the Licenses table. One provider can have many licenses.

Providers Table Fields

- ProviderId: The primary key for the provider.
- ProviderName: The name of the provider.
- County: The county location.
- Status: The current status of the provider.
- CreatedDate: The date when the record was created.
- IsDeleted: A flag that shows if the record is soft-deleted.
- DeletedAtUtc: The date and time of deletion in UTC.

Licenses Table Fields

- LicenseId: The primary key for the license.
- ProviderId: The foreign key that connects to the provider.
- LicenseNumber: The unique identifier of the license.
- LicenseStatus: The status of the license.
- ExpirationDate: The expiration date of the license.
- IsDeleted: A flag that shows if the record is soft-deleted.
- DeletedAtUtc: The date and time of deletion in UTC.

3. Soft Delete Functionality

The database uses a soft delete system to prevent data loss.

- The database does not remove records when a user deletes data.
- The system changes the IsDeleted field to true.
- The system writes the current UTC time into the DeletedAtUtc field.
- When a user deletes a provider, the system soft-deletes all related licenses automatically.
- Standard database queries do not show soft-deleted records.

4. API Endpoints

Providers Controller

Route base: `/api/providers`

Method	Endpoint	Description
GET	/api/providers	Gets a list of all active providers and their licenses.
POST	/api/providers	Creates a new provider record.
PUT	/api/providers/{id}	Updates an existing provider record.
DELETE	/api/providers/{id}	Soft deletes a provider and all connected licenses.
GET	/api/providers/repor…	Gets a report of active providers with valid licenses.
GET	/api/providers/repor…	Gets a report of active providers with expired licenses.
GET	/api/providers/audit…	Gets a list of soft-deleted providers for audit purposes.

Licenses Controller

Route base: /api/licenses

Method	Endpoint	Description
POST	/api/licenses	Creates a new license for a provider.
DELETE	/api/licenses/{id}	Soft deletes a single license.